

Auteur: Luca Letroye
Studierichting: Informatica
Oefeningenreeks 4A nr. 38.7

$$\begin{aligned} & \int \left(\frac{4x^2 + 4x + 1}{4x^2 + 4x + 3} \right) dx \\ &= \int \left(\frac{4x^2 + 4x + 3 - 2}{4x^2 + 4x + 3} \right) dx \\ &= \int \left(1 - \frac{2}{4x^2 + 4x + 3} \right) dx \\ &= \int \left(1 - \frac{2}{(2x + 1)^2 + 2} \right) dx \\ & t = 2x + 1 \Rightarrow dt = 2dx \\ & \Rightarrow \int \left(1 - \frac{1}{t^2 + 2} \right) dt \\ &= x - \frac{\sqrt{2}}{2} \operatorname{Bgtan} \left(\frac{2x + 2}{\sqrt{2}} \right) + C \end{aligned}$$

References